

Created by the Aerospace Cyber Defence Centre of Excellence (CEC)

Testing guide **WAC (Windows Artefacts Collector)**

Windows artefact collection software
for digital forensic investigation



SYSTEM
ARTEFACTS



USER
FILES



DIGITAL
EVIDENCE

A TOOL IN SUPPORT
OF DIGITAL
INVESTIGATION

WAC — Windows Artefacts Collector

Testing guide

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This document explains how to run each of the WAC tests by hand, what each one proves and does not prove, and how to read a failure. It is meant for whoever changes the code or wants to check a version before using it. The durations and volumes quoted were measured on the test virtual machine and on a real collection; they are orders of magnitude. The results of these tests, and what they guarantee, are in the *Validation report*.

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Principles

1.1 What a WAC test must prove

WAC produces JSON files. That they are valid proves almost nothing: a field read at the wrong offset, a date interpreted in the wrong time zone, a name decoded with the wrong code page all give perfectly valid JSON — and wrong JSON. The costliest defects found in WAC were all of that kind: they showed only when a value was confronted with another source.

The tests therefore fall into three families, which do not answer the same question:

The virtual machine cycle (chapter 3): does WAC run to the end on a real Windows, and are its outputs consistent with each other? The *cross-checks* confront a value computed by WAC with an independent value from the same collection.

The format harnesses (chapter 4): does a decoder give exactly the result of the reference — Microsoft's own tool, a published standard, another implementation?

The robustness harnesses (chapter 5): does a parser read outside its buffer, or loop for ever, on a truncated or forged file?

1.2 Three rules

1. **A fix comes with the test that would have caught the defect.** And that test is checked both ways: it passes on the fixed output, and it fails when the defect is replayed.
2. **After any change, compare with the previous run,** not just the green marks: an artefact that went from 1,522 entries down to 573 lights no warning at all (section 3.3).
3. **Real data does not leave the test workstation.** A real collection holds personal data: it is processed locally, its temporary copies are deleted after use, and none of it enters the repository (`vmtest/results/` is ignored by git).

Setting up

2.1 The Linux workstation

Everything is built on Linux, cross-compiled with MinGW-w64; the harnesses run natively, under wine, or in the virtual machine.

```
sudo apt install g++-mingw-w64-x86-64 wine64 python3 python3-pip
pip install olefile          # independent OLE reader, for the cross-checks
```

2.2 Building WAC and the harnesses

```
./build-windows.sh --test
```

Output, in build-windows/	Role
WAC.exe	the collection tool
raw_hive_test.exe	raw NTFS reader, used by the VM cycle
tests/lnk_test.exe	robustness of the shortcut parser
tests/parsers_test.exe	robustness of the jump list and Prefetch parsers
tests/evtx_test.exe, consigne_test.exe	event log decoder, exhibit store (chapter 4)
tests/offline_registry_test.exe	hive reader confronted with Microsoft's
tests/system_conversions_test.exe	WAC's conversions confronted with Windows'
tests/ntfs_fixup_test.exe	check of NTFS records (torn records)

Warning

The harnesses are deliberately in build-windows/tests/, away from WAC.exe: they are test tools, with no business on a collection medium. Never copy the tests/ folder onto one.

The build must end without a single warning: -Wall -Wextra are on, and warnings have already pointed at real defects.

The other harnesses (chapter 4) are each built with a command given in their header and repeated in this guide; several compile natively on Linux.

2.3 The test virtual machine

The VM is a Windows 11 driven through the *qemu-guest-agent*: no interaction, no network, commands run as NT AUTHORITY\SYSTEM. Its creation and prerequisites are described in `vmtest/README.md`; in short:

```
cd vmtest
ISO_WIN=~/Downloads/Win11_25H2_English_x64.iso ./create-vm.sh # once
python3 qga.py ping # does the agent answer?
python3 qga.py run --shell "dir C:\"" # run a command
python3 qga.py write <host> <guest> # send a file
python3 qga.py read <guest> <host> # fetch a file
```

Keep in mind

A command started with `qga.py run` is cut after 30 minutes. For a longer test (for instance the Prefetch files of a real machine, section 5.3), start it as a detached task in the VM and follow its log.

The test cycle in the virtual machine

3.1 Running a cycle

```
cd vmtest
./run-wac-test.sh --build      # rebuild, then full cycle
./run-wac-test.sh             # full cycle with the binaries already built
./run-wac-test.sh --raw-only  # only the validation of the raw reading
./run-wac-test.sh --no-split  # without step 7 (collection then conversion)
```

The cycle chains: upload of WAC.exe and raw_hive_test.exe into the VM, validation of the raw extraction of a hive (reg load must refuse it until it is repaired, then accept it), run of WAC, then fetch, into the folder vmtest/results/<timestamp>/, of:

- run.log — the console output of the collection, which ends with END, Time elapsed when WAC went to the end;
- raw.log — the trace of the NTFS reader;
- the JSON files of the artefacts;
- exhibits/ — the exhibit manifest, its seal and the list of the files really present in the exhibit store.

Step 7 then separates collection and conversion (chapter “Collect here, convert elsewhere” of the user documentation): --collect --events --binary, which must convert nothing; a conversion of an exhibit store with one retouched exhibit, which must be refused before any read, with its reason; a conversion, whose JSON files pass the checks below; a second conversion, whose JSON files must be identical to the first’s, byte for byte. Results in split/ and split-again/.

3.2 Checking the outputs

```
python3 check-json.py results/<timestamp>
```

Each line starts with a green tick (check passed), a red cross (defect) or a “skip” sign (check not applicable: the artefact is absent or empty). The exit code is 0 if no check failed. Beyond JSON validity and the shape of paths, the script confronts independent sources:

Cross-check	What it has already caught
Prefetch hash / suffix of its file name	65 wrong hashes out of 270 (hexadecimal without padding)
Prefetch executable name / start of the file name	a name read 8 bytes too early ("x11" everywhere)
Prefetch \VOLUME{guid} paths resolved	20,066 paths out of 20,066 unresolved
X / XUtc pairs, lists included	double time-zone shift; Prefetch RunsUtc 2 h off
offset of every local date / tz database	the collection day's offset applied to every date; daylight saving dates misread
no session earlier than the boot	date shifts, a boot time 3.5 s late
ShimCache date / NTFS date in the manifest	a UTC date treated as local (-2 h)
DestList entries / jump list shortcuts	the first 15 entries of every list lost
machine name of the events / OperatingSystem.json	check of the BinXML decoding
record identifiers unique per log	a stale chunk read again
mounts / reg query and Get-Partition	device paths published as "\"
precision of FAT and text dates	seven digits of fraction on dates to the second
BAM, UserAssist, USBSTOR, Amcache / harness, Prefetch, Windows, PE headers	UTC dates read as local (2 h)
SID names / Get-LocalUser, Get-LocalGroup	names asked of the system, translated ("Système")
GUIDs registered by the harness (register-test-guids.ps1)	"Unmapped GUID" published as a name
seal / real fingerprint of the manifest	a retouched manifest
files in the exhibit store / manifest content	209 exhibits added after sealing
SHA-256 of files read raw / Get-FileHash	safeguard of the reading by batches of clusters
Prefetch dates / dates given by Windows	0 out of 335 matching: dates of the working copy
executables of System32 and SysWOW64 / manifest of a --collect --binary	6,446 missing out of 8,540 (\$INDEX_ROOT in an extension record, hard links)
catalogs cited in a verdict / exhibits of the store	170 out of 170 missing (label "catalog" looked for as "catalogue")

Cross-check	What it has already caught
WAC's Microsoft verdicts / Get-AuthenticodeSignature	safeguard: a wrong verdict would leave a binary out of the store
copy of a Store package (untouched, modified, added) / verdicts	verification through the signed block map
two conversions of one collection	dates read on the working copy, rebuilt at each conversion

The full list, with the history of each check, is in `vmtest/README.md`.

3.3 Comparing two runs

A green mark does not tell whether a piece of data disappeared. After any change, compare the new collection with the previous one: number of entries of each artefact, and values lost or gained. Two precautions:

- a collection changes by itself from one run to the next: timestamps, new events, updated Prefetch files, the fingerprint of \$MFT. Compare *paired* items — for instance a Prefetch by its path hash and its last run — leaving aside the dates of the extracted copy;
- a value that looks “lost” may just be an empty field no longer emitted, or a renamed key: normalise before concluding.

CHAPTER 4

The format harnesses

Each one confronts a WAC decoder with a reference. All are excluded from the WAC build by their `_test.cpp` suffix.

4.1 Fingerprints: `sha_test`

Confronts SHA-1 and SHA-256 with the published vectors of FIPS 180-4, and with lengths 55 to 128 bytes, which exercise the padding — the one place where an otherwise correct implementation goes wrong.

```
cd WAC && g++ -std=c++17 -I. sha.cpp sha_test.cpp -o /tmp/sha_test && /tmp/sha_test
```

4.2 JSON reading: `json_test`

`--convert` reads back what `--collect` wrote: the exhibit manifest and the snapshot of the live state. The harness checks that everything WAC's writer produces reads back identical, that the files of a real collection read back and rewrite to the byte, that malformed texts are refused, and that 200,000 random mutations never make the reader read out of bounds (AddressSanitizer, UBSan).

```
cd WAC && g++ -std=c++17 -g -fsanitize=address,undefined -I. json_test.cpp -o /tmp/  
    json_test  
/tmp/json_test ../vmtest/results/<timestamp>/*.json
```

4.3 Decompression: `lznt1_test` and `xpress_test`

Check the NTFS (LZNT1) and WOF (XPRESS Huffman) decompressors against *Microsoft's own compressor*: a known file is compressed in the VM by `RtlCompressBuffer`, unit by unit, and the harness decompresses each unit and compares it byte for byte with the original. A wrong decompressor usually decodes without any error: only the exhaustive comparison reveals it.

```
cd WAC  
g++ -std=c++17 -I. lznt1.cpp lznt1_test.cpp -o /tmp/lznt1_test  
g++ -std=c++17 -I. xpress.cpp xpress_test.cpp -o /tmp/xpress_test  
/tmp/lznt1_test <original> <compressed.bin> <index.idx>  
/tmp/xpress_test <original> <compressed.bin> <index.idx>
```

index.idx holds one line <n> <original size> <compressed size> per unit; a null compressed size marks a unit Windows gave up compressing.

4.4 Event messages: wevt_test and event_messages_test

wevt_test checks the chain that turns an event into a sentence, on the files of a real provider: its DLL and its localised satellite.

```
cd WAC && g++ -std=c++17 -I. pe_resource.cpp wevt.cpp wevt_test.cpp -o /tmp/wevt_test
/tmp/wevt_test gpsvc.dll fr-FR/gpsvc.dll.mui "{aeal4fa-97d1-45f2-a64c-4d69fffd92c9}" 1002
4001
```

event_messages_test does the same starting from the SOFTWARE hive; it runs on the examined machine or in the VM, the only places where the hive and the providers' binaries are readable:

```
event_messages_test <SOFTWARE hive> <guid> [id[:version] ...]
```

4.5 Event logs: evtx_test

Reads an .evtx outside Windows, under wine; built by build-windows.sh --test, in build-windows/tests/. It serves to try the decoder on logs the VM does not produce: old, forwarded, closed abruptly, damaged — and to compare record by record with an independent implementation (python-evtx).

```
wine evtx_test.exe "Z:/path/Security.evtx"          # summary per log
wine evtx_test.exe "Z:/path/Security.evtx" 3        # + XML of 3 records
wine evtx_test.exe "Z:/path/Security.evtx" --dump    # one record per line
wine evtx_test.exe --collect "Z:/tmp/tree" "Z:/tmp/out"      # complete chain
wine evtx_test.exe --collect-memory "Z:/tmp/tree" "Z:/tmp/out2" # same chain, in memory
cmp /tmp/out/events.json /tmp/out2/events.json          # must be silent
```

4.6 Hive reader: offline_registry_test

WAC reads registry hives with its own reader (offline_registry.cpp), linked into the executable: it no longer depends on Microsoft's offreg.dll, which Windows ships and which it therefore borrowed from the examined machine. This harness confronts the two: it loads Microsoft's DLL from the path given, walks each hive with both readers and compares every key (name, date, class, counts, maxima, security descriptor size) and every value (name, type, bytes), as well as the contract's edge cases: buffer too small, size probe, default value, missing key or value. No difference is allowed.

```
offline_registry_test.exe C:\Windows\System32\offreg.dll SYSTEM SOFTWARE SAM ntuser.dat
...
```

To be run on Windows (the VM), on replayed hives: those of a collection's working/ folder. The one intended difference: a hive whose logs were not replayed is refused by WAC, where recent versions of Microsoft's DLL read it as is — stale keys, without an error; the harness reports it as expected. Reference: 18 hives, ten of them from a real machine, i.e. 1,273,652 keys and 2,456,139 values, identical.

4.7 Conversions: `system_conversions_test`

WAC imports only what querying the running system requires (ADVAPI32, KERNEL32, msvcrt, Secur32, WTSAPI32). The conversions it used to ask of other libraries are done by its own code; this harness confronts them with the Windows functions they replaced: GUIDs (StringFromGUID2), SIDs both ways (ConvertSidToStringSidW, ConvertStringSidToSidW), OLE dates (VariantTimeToSystemTime) and property names (PSGetNameFromPropertyKey), FAT dates (DosDateTimeToFileTime, on every date), and daylight saving time: WAC's offset in both directions (SystemTimeToTzSpecificLocalTimeEx, TzSpecificLocalTimeToSystemTimeEx) on *every* time zone of Windows, with its rules year by year, at the transitions (hour skipped, hour repeated, change of rule on 1 January) from 1980 to 2035; the reading of SYSTEM\...\TimeZoneInformation, whose daylight saving dates are TIME_FIELDS, not SYSTEMTIMES; rules forged at random; the reading back of the manifest's ISO 8601 dates (iso8601UtcToFiletime, exact to the 100 ns on 200,000 instants, malformed dates refused). It also checks the suspect's offset in both directions: a local date converted to UTC must be so with the time zone of the examined machine, not that of the machine running WAC. An impossible date (month 13, hour 25) gives a null date, hence not emitted. The table of property names is generated from the schema of a reference Windows by `vmtest/property-schema.cpp`.

```
system_conversions_test.exe
```

To be run on Windows (the VM). Reference: 9,861,570 comparisons, no difference. Two intended differences: WAC refuses a SID text Windows would truncate (a sub-authority of 2^{32} or more, a sixteenth sub-authority), and a NaN date, for which Windows returns a meaningless time.

4.8 Exhibit store: `consigne_test`

Replays the production chain — fingerprints, manifest, verified copy, hive replay — on hives given as arguments, and checks what no build reveals: the exhibit store stays identical byte for byte while the working copy is changed, and the seal carries the manifest's real fingerprint. Last, it checks that an exhibit path read from a manifest stays inside the exhibit store: a .. component is refused, a WinSxS name holding .. (i.e. `sermode`) is accepted.

```
cd build-windows/tests
wine consigne_test.exe "Z:/tmp/out" "Z:/path/SYSTEM" "Z:/path/ntuser.dat"
```

4.9 Raw reading: `raw_hive_test`

Extracts a file by reading the NTFS volume directly. To be run on Windows, as an administrator; the VM cycle uses it to validate the extraction of a hive.

```
raw_hive_test.exe [volume] [path] [output] [--fix] [--list] [--attrs]
```

`--fix` makes the extracted hive loadable, `--list` lists a directory, `--attrs` shows the \$MFT attributes of the file without interpreting them.

4.10 NTFS records: `ntfs_fixup_test`

NTFS writes a \$MFT record or an index block sector by sector, and ends every 512-byte stride with a sequence number. WAC checks that number before restoring the real bytes: a record torn by an interrupted write is refused. The harness builds records whose expected result is known (sound, torn in each stride, malformed arrays), then confronts WAC's verdict with an independent check on 20,000 random records placed against a forbidden page. No data is needed.

```
cd build-windows/tests && wine ntfs_fixup_test.exe
```

Reference: 44,610 checks, no failure. On a real volume, the VM cycle confirms it: `raw.log` must hold no "torn record".

4.11 Signatures: `authenticode_test`

Verifies the catalogs of a folder, then gives a verdict per file; its verdicts are compared with those of `Get-AuthenticodeSignature` on the same files, in the VM. The only gap that matters: a file WAC accepts and Windows refuses.

```
authenticode_test --catalogs <folder>  
authenticode_test --catalogs <folder> --files <list>
```

The robustness harnesses

5.1 The principle

A parser reads at offsets the file itself declares. On a truncated or forged file, a read past the buffer does not crash on its own: it reads the neighbouring memory and publishes it, in valid JSON. The harnesses make it crash: each input is copied against a forbidden memory page (PAGE_NOACCESS), once *ending* against it (reads past the end), once *starting* right after it (reads before the start, for instance at a negative offset taken from the data). A single byte read outside the input raises an access violation.

For each file given, the harness parses:

- the whole file;
- its truncations (every length for small files);
- copies with bytes overwritten at random, with a fixed seed: a failure reproduces identically.

5.2 Shortcuts: lnk_test

```
cd build-windows/tests
wine lnk_test.exe a.lnk b.lnk ...
```

Where to find files: the Recent folder and the Start menu of the VM, or those of a collection (Users\<name>\AppData\Roaming\Microsoft\Windows\Recent). Accented or non-Latin names travel badly through the wine command line: copy the files under numbered names.

5.3 Jump lists and Prefetch: parsers_test

```
parsers_test.exe jumplist-auto <file.automaticDestinations-ms> ...
parsers_test.exe jumplist-custom <file.customDestinations-ms> ...
parsers_test.exe prefetch <file.pf> ...
parsers_test.exe hive <hive> ...
```

The hive kind tries the hive reader: each input is opened from memory, against the forbidden page, and the whole tree is walked (subkeys, values, ORQueryInfoKey). A subkey pointing back at one of its ancestors is refused by the reader: without that, a recursive walk would never end.

Jump lists are tested under wine, like `lnk_test`. Prefetch files are tested **in the VM**: they are compressed (XPRESS Huffman), and only the `ntdll` library of Windows decompresses them. The harness decompresses each file before trying it: it is WAC's parser it aims at, not Windows' decompressor.

For the 307 Prefetch files of a real machine (over a million inputs), count about two hours in the VM: run the test as a detached task, in batches, and follow its log. Example PowerShell script, put into the VM with `qga.py write`:

```
$files = Get-ChildItem C:\wactest\pt\pf\*.pf | ForEach-Object { $_.FullName }
$log = 'C:\wactest\pt\pf_results.txt'
for ($i = 0; $i -lt $files.Count; $i += 20) {
    $batch = $files[$i..([Math]::Min($i + 19, $files.Count - 1))]
    Add-Content $log (& C:\wactest\pt\parsers_test.exe prefetch @batch 2>&1)
    Add-Content $log ("batch " + $i + " exit=" + $LASTEXITCODE)
}
Add-Content $log 'DONE'
```

5.4 Reference

Results obtained on a real machine, each input parsed on both sides of the forbidden page:

Files	Inputs	Where
220 shortcuts	546,214	wine
48 automatic jump lists	175,585	wine
23 custom jump lists	79,036	wine
307 Prefetch files	1,246,491	VM, about 2 h
4 registry hives	16,207	wine

Cross-checking on a real collection

A freshly installed VM produces no fragmentation, no old jump lists, no exotic file names. Several defects only showed on a real collection, confronted with an independent source:

- **jump lists / olefile:** the streams and the DestList, read again by an independent OLE reader, must give exactly the same entries (numbers, dates, paths). That is what showed the streams are named "1" to "f", not "01" to "0f";
- **shortcut / DestList:** in a jump list, the target of a shortcut and the path of its DestList entry are two records of the same file. Their disagreement revealed a wrong code page ("d\x92orientation") and an ignored Unicode path;
- **events / python-evtx:** record-by-record comparison (evtx_test --dump);
- **signatures / Get-AuthenticodeSignature** (authenticode_test);
- **old collection / new one:** decode the same files with the current code and compare, after normalisation, with the output of a previous version.

Warning

A real collection is processed locally. If a hive must be confronted with Microsoft's reader (offline_registry_test), it may be copied into the test VM, then deleted as soon as the test is over. No copy stays in the temporary working folder, and none of it enters the repository.

Reading a failure

7.1 Exit codes

Code	Meaning
0	everything passed
1	a check failed, or an input file is unreadable
2	wrong usage (arguments)
3	parsers_test: a parse did not end within 60 s (watchdog)
0xC0000005 (-1073741819)	access violation: a read outside the buffer

7.2 Locating a read outside the buffer

Under wine, a fault prints the call stack, with addresses relative to the program (lnk_test+0x485bb). On Windows, parsers_test itself prints the input at fault and the faulting address:

```
FAILED exception 0xC0000005 at exe+0x78b70, address read 0x159deb4200c:
C:\wactest\pt\pf\CMD.EXE-0BD30981.pf truncated at 192 (guard after)
```

The address is translated into a function with the program's symbol table (base address 0x140000000):

```
x86_64-w64-mingw32-nm -C parsers_test.exe | grep -i " t " | sort > syms.txt
# look for the symbol whose address is just below 0x140000000 + 0x78b70
x86_64-w64-mingw32-objdump -d --start-address=0x140078b40 \
--stop-address=0x140078b80 parsers_test.exe
```

The disassembled instruction gives the offset read (here 0xcc(%r14)); it is then matched, in the source code, with the field read at that position. Then replay the exact input (the truncation or corruption shown) to check the fix, and the whole series after it.

7.3 An endless loop

If parsers_test stops on “no end after 60 s”, the input named keeps a parser running for ever — typically a cyclic sector chain in an OLE container. It is a defect just like a read outside

the buffer: a forged file would be enough to block a collection.